STA 209: Introduction to Time Series Analysis

Homework 2

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February 19, 2026

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Disclosure

GPT-5.3-Codex was used to create the `YAML` portion and some `LaTeX` code to format the text/equations nicely. Page formatting code was also provided by Derin Gezgin. In the setup chunk, libraries were loaded and some helper functions were defined including but not limited to `table_latex()` and `with_family()`. See the original `RMD` file here for more details.

Problem 1

Problem 1

Consider the Southern oscillation index data (`soi`) from Homework 1. Use this data to do the following in R and report your results. Use periodicity $s = 12$. Fit the following five models:

$$
\begin{aligned}
\textbf{Model I:} \quad & \mathrm{SOI}_t = \beta_0 + \beta_1 t + W_t, \\
\textbf{Model II:} \quad & \mathrm{SOI}_t = \beta_0 + \beta_1 t + \frac{\beta_2 t^2}{2!} + W_t, \\
\textbf{Model III:} \quad & \mathrm{SOI}_t = \beta_0 + \alpha_1 \cos\left(\frac{2\pi t}{s}\right) + \alpha_2 \sin\left(\frac{2\pi t}{s}\right) + W_t, \\
\textbf{Model IV:} \quad & \mathrm{SOI}_t = \beta_0 + \beta_1 t + \alpha_1 \cos\left(\frac{2\pi t}{s}\right) + \alpha_2 \sin\left(\frac{2\pi t}{s}\right) + W_t, \\
\textbf{Model V:} \quad & \mathrm{SOI}_t = \beta_0 + \beta_1 t + \alpha_1 \cos\left(\frac{2\pi t}{s}\right) + \alpha_2 \sin\left(\frac{2\pi t}{s}\right) \\
& \qquad + \alpha_3 \cos\left(\frac{4\pi t}{s}\right) \\
& \qquad + \alpha_4 \sin\left(\frac{4\pi t}{s}\right) \\
& \qquad + W_t.
\end{aligned}
$$

The Southern Oscillation Index (SOI) dataset is a monthly time series with 453 observations covering the years 1950 through 1987. It is represented in the form `Time-Series [1:453]` (notated as running from 1950 to 1988 in the printed format) and is shown as a numeric sequence (e.g., `0.377, 0.246, 0.311, 0.104, -0.016, 0.235, ...`). The series is intended to pair with `rec` (Recruitment) for related analyses. The data were furnished by Dr. Roy Mendelssohn of the Pacific Fisheries Environmental Laboratory, NOAA (personal communication). Demonstrations of `astsa` capabilities are referenced under *FUN WITH ASTSA*, and the most recent version of the package is maintained on GitHub along with the `NEWS` and `ChangeLog` files; additional webpages for the associated texts and help using `R` for time series analysis are available at the author's site.

Problem 1 (a)

Problem 1 (a)

Report the five fitted models.

Cell 1

```r
# Load SOI data and define periodicity.
data(soi)
s <- 12
Y <- soi
t <- time(Y)
ts2 <- t^2 / factorial(2)
cs1 <- cos(2 * pi * t / s)
sn1 <- sin(2 * pi * t / s)
cs2 <- cos(4 * pi * t / s)
sn2 <- sin(4 * pi * t / s)

# Fit all five models.
m1 <- lm(Y ~ t)
m2 <- lm(Y ~ t + ts2)
m3 <- lm(Y ~ cs1 + sn1)
m4 <- lm(Y ~ t + cs1 + sn1)
m5 <- lm(Y ~ t + cs1 + sn1 + cs2 + sn2)

# See model summaries.
summary(m1)
```

```
##
## Call:
## lm(formula = Y ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.04140 -0.24183  0.01935  0.27727  0.83866
##
## Coefficients:
##             Estimate Std. Error t value  Pr(>|t|)
## (Intercept) 13.70367    3.18873   4.298 0.0000212 ***
## t           -0.00692    0.00162  -4.272 0.0000236 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3756 on 451 degrees of freedom
## Multiple R-squared:  0.0389, Adjusted R-squared:  0.03677
## F-statistic: 18.25 on 1 and 451 DF,  p-value: 0.00002359
```

Cell 2

```r
summary(m2)
```

```
##
## Call:
## lm(formula = Y ~ t + ts2)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -1.0529 -0.2443  0.0129  0.2658  0.8412
##
## Coefficients:
##                 Estimate   Std. Error t value Pr(>|t|)
## (Intercept) -496.7608959  644.3532809  -0.771    0.441
## t              0.5116415    0.6545674   0.782    0.435
## ts2           -0.0002634    0.0003325  -0.792    0.429
##
## Residual standard error: 0.3758 on 450 degrees of freedom
## Multiple R-squared:  0.04024,    Adjusted R-squared:  0.03597
## F-statistic: 9.433 on 2 and 450 DF,  p-value: 0.00009698
```

Cell 3

```r
summary(m3)
```

```
##
## Call:
## lm(formula = Y ~ cs1 + sn1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08705 -0.26231  0.03387  0.28045  0.93815
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)  0.0794096  0.0180282   4.405 0.0000133 ***
## cs1         -0.0008964  0.0251756  -0.036     0.972
## sn1         -0.0312776  0.0258020  -1.212     0.226
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.383 on 450 degrees of freedom
## Multiple R-squared:  0.003266,   Adjusted R-squared:  -0.001164
## F-statistic: 0.7373 on 2 and 450 DF,  p-value: 0.479
```

Cell 4

```r
summary(m4)
```

```
##
## Call:
## lm(formula = Y ~ t + cs1 + sn1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0538 -0.2431  0.0269  0.2770  0.8546
##
## Coefficients:
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) 13.575646   3.271475   4.150 0.0000398 ***
## t           -0.006855   0.001662  -4.125 0.0000441 ***
## cs1         -0.010721   0.024854  -0.431     0.666
## sn1         -0.010208   0.025864  -0.395     0.693
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3763 on 449 degrees of freedom
## Multiple R-squared:  0.03967,    Adjusted R-squared:  0.03325
## F-statistic: 6.182 on 3 and 449 DF,  p-value: 0.0004011
```

Cell 5

```r
summary(m5)
```

```
##
## Call:
## lm(formula = Y ~ t + cs1 + sn1 + cs2 + sn2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.06440 -0.24486  0.01442  0.27100  0.84117
##
## Coefficients:
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) 12.797341   3.238141   3.952 0.0000900 ***
## t           -0.006460   0.001645  -3.928 0.0000992 ***
## cs1         -0.008360   0.024591  -0.340    0.7340
## sn1         -0.009441   0.025548  -0.370    0.7119
## cs2         -0.043304   0.024829  -1.744    0.0818 .
## sn2          0.081218   0.024730   3.284    0.0011 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3716 on 447 degrees of freedom
## Multiple R-squared:  0.06798,    Adjusted R-squared:  0.05755
## F-statistic:  6.52 on 5 and 447 DF,  p-value: 0.000007219
```

The fitted models are:

$$\widehat{\textbf{Model I:}} \quad \widehat{\mathrm{SOI}}_t = 13.704 - 0.007\,t,$$

$$\widehat{\textbf{Model II:}} \quad \widehat{\mathrm{SOI}}_t = -496.761 + 0.512\,t - 0.000\,\frac{t^2}{2!},$$

$$\widehat{\textbf{Model III:}} \quad \widehat{\mathrm{SOI}}_t = 0.079 - 0.001\,\cos\left(\frac{2\pi t}{12}\right) - 0.031\,\sin\left(\frac{2\pi t}{12}\right),$$

$$\widehat{\textbf{Model IV:}} \quad \widehat{\mathrm{SOI}}_t = 13.576 - 0.007\,t - 0.011\,\cos\left(\frac{2\pi t}{12}\right) - 0.010\,\sin\left(\frac{2\pi t}{12}\right),$$

$$\widehat{\textbf{Model V:}} \quad \widehat{\mathrm{SOI}}_t = 12.797 - 0.006\,t - 0.008\,\cos\left(\frac{2\pi t}{12}\right) - 0.009\,\sin\left(\frac{2\pi t}{12}\right) - 0.043\,\cos\left(\frac{4\pi t}{12}\right) + 0.081\,\sin\left(\frac{4\pi t}{12}\right).$$

where SOI_t denotes the Southern Oscillation Index at time t, and $s = 12$ is the monthly seasonal period.

Problem 1 (b)

Problem 1 (b)

Prepare a summary table using number of regression parameters, R_a^2, AIC, AICC and BIC.

Cell 6

```r
# Build table of fit criteria.
fit_table <- function(fit) {
  n   <- length(resid(fit))
  r   <- length(coef(fit))
  sse <- sum(resid(fit)^2)
  data.frame(
    Parameters = r,
    R2a  = summary(fit)$adj.r.squared,
    AIC  = AIC(fit) / n,
    AICC = log(sse / n) + ((n + r) / (n - r - 2)),
    BIC  = AIC(fit, k = log(n)) / n
  )
}
comp1 <- dplyr::bind_rows(
  'Model I'   = fit_table(m1),
  'Model II'  = fit_table(m2),
  'Model III' = fit_table(m3),
  'Model IV'  = fit_table(m4),
  'Model V'   = fit_table(m5),
  .id = 'Model'
)
comp1$Parameters <- as.integer(comp1$Parameters)
comp1 <- comp1 %>%
  dplyr::mutate(
    R2a  = round(R2a, 3),
    AIC  = round(AIC, 3),
    AICC = round(AICC, 3),
    BIC  = round(BIC, 3)
  )
table_latex(comp1, col_names = c('Model', '\\# Parameters', '$R_a^2$', 'AIC',
'AICC', 'BIC'))
```

Model	# Parameters	R_a^2	AIC	AICC	BIC
Model I	2	0.037	0.888	-0.949	0.916
Model II	3	0.036	0.891	-0.946	0.928
Model III	3	-0.001	0.929	-0.908	0.966
Model IV	4	0.033	0.897	-0.941	0.942
Model V	6	0.058	0.875	-0.962	0.939

Problem 1 (c)

Problem 1 (c)

Using the summary in (b), which model is selected to explain SOI and why?

For explanation, we prioritize adjusted R^2 and the overall model-summary F-tests. Model V has the largest adjusted R^2, with $R_a^2 = 0.058$. Its overall F-test is significant with $F = 6.520$ and $p = 0.000$. Also, Model III has $F = 0.737$ and $p = 0.479$, so it is not significant. Therefore, **Model V** is selected for explaining SOI variation among these candidates.

Problem 1 (d)

Problem 1 (d)

Based on AIC, AICC and BIC, which model is selected and what is it best for?

AIC and AICC are minimized by **Model V**, with $AIC = 0.875$ and $AICC = -0.962$, but BIC is minimized by **Model I**, with $BIC = 0.916$. Since these criteria disagree and BIC applies the strongest complexity penalty, we select **Model I**. This selected model is best for a more parsimonious in-sample representation rather than maximizing seasonal flexibility.

Problem 1 (e)

Problem 1 (e)

Remove the last 30 values and compare future forecast error across models using ME, MPE, MSE, MAE, and MAPE. Include these with R_a^2, AIC, AICC and BIC in a table. Does the selected model match part (c)?

Cell 7

```r
# Hold out the last 30 observations.
h     <- 30
n_all <- length(Y)
cut   <- n_all - h
Y_train   <- stats::window(Y, end = time(Y)[cut])
Y_test    <- stats::window(Y, start = time(Y)[cut + 1])
t_train   <- time(Y_train)
t_test    <- time(Y_test)
ts2_train <- t_train^2 / factorial(2)
ts2_test  <- t_test^2 / factorial(2)
cs1_train <- cos(2 * pi * t_train / s)
sn1_train <- sin(2 * pi * t_train / s)
cs2_train <- cos(4 * pi * t_train / s)
sn2_train <- sin(4 * pi * t_train / s)
cs1_test  <- cos(2 * pi * t_test / s)
sn1_test  <- sin(2 * pi * t_test / s)
cs2_test  <- cos(4 * pi * t_test / s)
sn2_test  <- sin(4 * pi * t_test / s)

# Refit on training data.
f1 <- lm(Y_train ~ t_train)
f2 <- lm(Y_train ~ t_train + ts2_train)
f3 <- lm(Y_train ~ cs1_train + sn1_train)
f4 <- lm(Y_train ~ t_train + cs1_train + sn1_train)
f5 <- lm(Y_train ~ t_train + cs1_train + sn1_train + cs2_train + sn2_train)

# Forecast the held-out 30 observations.
p1 <- predict(f1, newdata = data.frame(t_train = t_test))
p2 <- predict(f2, newdata = data.frame(t_train = t_test, ts2_train = ts2_test))
p3 <- predict(f3, newdata = data.frame(cs1_train = cs1_test, sn1_train =
sn1_test))
p4 <- predict(f4, newdata = data.frame(t_train = t_test, cs1_train = cs1_test,
sn1_train = sn1_test))
p5 <- predict(f5, newdata = data.frame(t_train = t_test, cs1_train = cs1_test,
sn1_train = sn1_test, cs2_train = cs2_test, sn2_train = sn2_test))

# Forecast error metrics (Lecture 5 style).
fcst_metrics <- function(y_true, y_hat) {
  e <- y_true - y_hat
  data.frame(
    ME = mean(e),
    MPE = 100 * mean(e / y_true),
    MSE = mean(e^2),
```

```r
    MAE = mean(abs(e)),
    MAPE = 100 * mean(abs(e / y_true))
  )
}
fc_table <- dplyr::bind_rows(
  'Model I' = fcst_metrics(Y_test, p1),
  'Model II' = fcst_metrics(Y_test, p2),
  'Model III' = fcst_metrics(Y_test, p3),
  'Model IV' = fcst_metrics(Y_test, p4),
  'Model V' = fcst_metrics(Y_test, p5),
  .id = 'Model'
)

# Training-fit criteria for the same refit models.
comp_train <- dplyr::bind_rows(
  'Model I' = fit_table(f1),
  'Model II' = fit_table(f2),
  'Model III' = fit_table(f3),
  'Model IV' = fit_table(f4),
  'Model V' = fit_table(f5),
  .id = 'Model'
)

# Round fit criteria table.
comp_fit <- comp_train %>%
  dplyr::mutate(
    Parameters = as.integer(Parameters),
    R2a = round(R2a, 3),
    AIC = round(AIC, 3),
    AICC = round(AICC, 3),
    BIC = round(BIC, 3)
  )

# Round forecast error table.
comp_fc <- fc_table %>%
  dplyr::mutate(
    ME = round(ME, 3),
    MPE = round(MPE, 3),
    MSE = round(MSE, 3),
    MAE = round(MAE, 3),
    MAPE = round(MAPE, 3)
  )

# Print two tables.
table_latex(
  comp_fit,
  col_names = c('Model', '\\# Parameters', '$R_a^2$', 'AIC', 'AICC', 'BIC')
)
```

Model	# Parameters	R_a^2	AIC	AICC	BIC
Model I	2	0.038	0.903	-0.935	0.932
Model II	3	0.041	0.902	-0.936	0.940
Model III	3	-0.001	0.945	-0.893	0.983
Model IV	4	0.033	0.912	-0.925	0.960
Model V	6	0.055	0.894	-0.943	0.961

Cell 8

```
table_latex(
  comp_fc,
  col_names = c('Model', 'ME', 'MPE', 'MSE', 'MAE', 'MAPE')
)
```

Model	ME	MPE	MSE	MAE	MAPE
Model I	0.064	105.563	0.115	0.300	105.867
Model II	0.142	113.126	0.132	0.315	125.284
Model III	-0.097	90.121	0.117	0.292	100.952
Model IV	0.057	104.894	0.114	0.299	104.894
Model V	0.057	101.014	0.106	0.293	110.106

Using future-forecast error, **Model V** is preferred by MSE, with $MSE = 0.106$. MAE and MAPE are slightly smaller for Model III, so the error criteria are not fully unanimous; however, with MSE as the main forecasting criterion, Model V is selected. Therefore, the best model in part (e) matches part (c), but does not match part (d).

Problem 2

Problem 2 (30 points)

For the New Haven Average Temperature time series, fit the following trend and seasonality models:

$$\textbf{Model I:} \quad \text{AvgTemp}_t = \beta_0 + \beta_1 t + \alpha_1 \cos\left(\frac{2\pi t}{s}\right) + \alpha_2 \sin\left(\frac{2\pi t}{s}\right) + W_t,$$

$$\textbf{Model II:} \quad \text{AvgTemp}_t = \beta_0 + \beta_1 t + \frac{\beta_2 t^2}{2!} + \alpha_1 \cos\left(\frac{2\pi t}{s}\right) + \alpha_2 \sin\left(\frac{2\pi t}{s}\right) + W_t,$$

$$\textbf{Model III:} \quad \text{AvgTemp}_t = \beta_0 + \beta_1 t + \frac{\beta_2 t^2}{2!} + \alpha_1 \cos\left(\frac{2\pi t}{s}\right) + \alpha_2 \sin\left(\frac{2\pi t}{s}\right) + \alpha_3 \cos\left(\frac{4\pi t}{s}\right) + \alpha_4 \sin\left(\frac{4\pi t}{s}\right) + W_t.$$

Problem 2 (a)

Problem 2 (a)

Determine s using periodogram.

Cell 9

```r
# Load New Haven average temperature series.
NewHaven <- ts_seasonal('data/Climate_Northeast_NewHavenCT.xlsx')
Y <- make_site_ts(NewHaven)$ts_avg

# Periodogram (Lecture 6 style).
x <- Y - mean(Y)
n <- length(x)
I <- abs(fft(x))^2 / n
half <- floor((n / 2) - 1)
f <- 0:half / n
P <- (4 / n) * I[1:(n / 2)]
index <- which(P == max(P))
s <- round(1 / f[index])

# Report peak frequency and period.
f[index]
```

```
## [1] 0.25
```

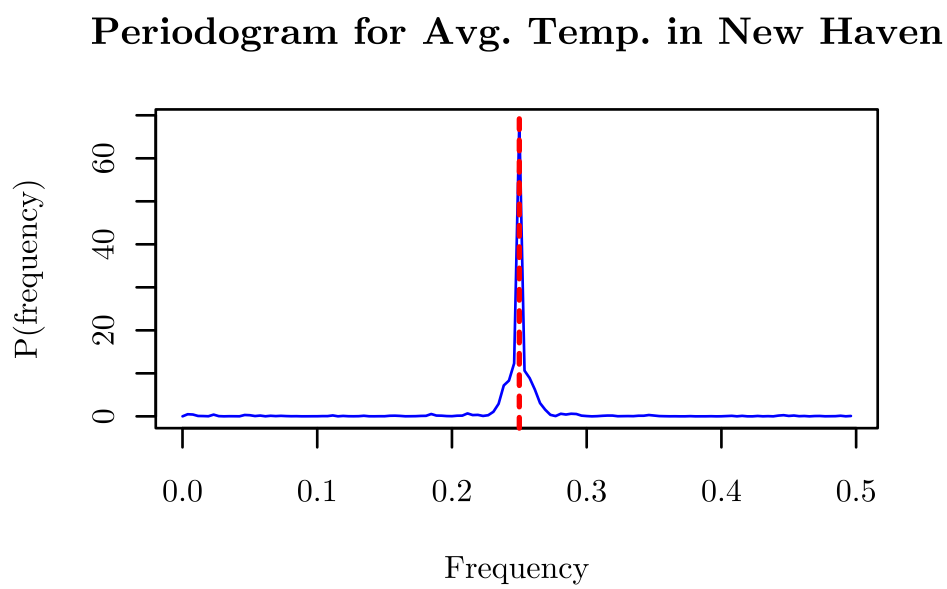
Cell 10

```r
s
```

```
## [1] 4
```

Cell 11

```r
# Plot periodogram.
with_family({
  plot(f, P, type = 'l',
       main = 'Periodogram for Avg. Temp. in New Haven',
       xlab = 'Frequency', ylab = 'P(frequency)', col = 'blue')
  abline(v = f[index], col = 'red', lty = 2, lwd = 2)
})
```



The periodogram has its dominant peak at frequency $f = 0.25$, so the estimated seasonal period is

$$s = \frac{1}{0.25} = 4.$$

Thus, the New Haven average temperature series is modeled with quarterly seasonality ($s = 4$).

Problem 2 (b)

Problem 2 (b)

Report the three fitted models.

Cell 12

```r
# Build regressors and fit the three models.
t    <- time(Y)
t2   <- t^2 / factorial(2)
cs1  <- cos(2 * pi * t / s)
sn1  <- sin(2 * pi * t / s)
cs2  <- cos(4 * pi * t / s)
sn2  <- sin(4 * pi * t / s)
fit1 <- lm(Y ~ t + cs1 + sn1)
fit2 <- lm(Y ~ t + t2 + cs1 + sn1)
fit3 <- lm(Y ~ t + t2 + cs1 + sn1 + cs2 + sn2)

# See model summaries.
summary(fit1)
```

```
##
## Call:
## lm(formula = Y ~ t + cs1 + sn1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.262  -7.521  -0.115   4.648  14.960
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.232756  56.200355   0.075    0.940
## t            0.003537   0.028349   0.125    0.901
## cs1         -0.044970   0.750657  -0.060    0.952
## sn1         -0.095639   0.753658  -0.127    0.899
##
## Residual standard error: 8.572 on 256 degrees of freedom
## Multiple R-squared:  0.000136,   Adjusted R-squared:  -0.01158
## F-statistic: 0.01161 on 3 and 256 DF,  p-value: 0.9983
```

Cell 13

```r
summary(fit2)
```

```
##
## Call:
## lm(formula = Y ~ t + t2 + cs1 + sn1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.2258  -7.0674   0.2296   4.2398  14.1111
##
## Coefficients:
##                Estimate  Std. Error t value Pr(>|t|)
## (Intercept) 5952.733985 6643.803267   0.896    0.371
## t             -5.998389    6.703292  -0.895    0.372
## t2             0.003028    0.003381   0.895    0.371
```

```
## cs1           -0.019103    0.751504  -0.025    0.980
## sn1           -0.078355    0.754197  -0.104    0.917
##
## Residual standard error: 8.576 on 255 degrees of freedom
## Multiple R-squared:  0.00327,    Adjusted R-squared:  -0.01237
## F-statistic: 0.2091 on 4 and 255 DF,  p-value: 0.9332
```

Cell 14

```r
summary(fit3)
```

```
##
## Call:
## lm(formula = Y ~ t + t2 + cs1 + sn1 + cs2 + sn2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.2611  -7.1015   0.2688   4.2093  13.9779
##
## Coefficients:
##                Estimate  Std. Error t value Pr(>|t|)
## (Intercept) 5914.416992 6672.746965   0.886    0.376
## t             -5.959728    6.732495  -0.885    0.377
## t2             0.003008    0.003396   0.886    0.377
## cs1           -0.017235    0.754501  -0.023    0.982
## sn1           -0.076877    0.757175  -0.102    0.919
## cs2            0.034329    0.755150   0.045    0.964
## sn2            0.137476    0.755529   0.182    0.856
##
## Residual standard error: 8.609 on 253 degrees of freedom
## Multiple R-squared:  0.003408,   Adjusted R-squared:  -0.02023
## F-statistic: 0.1442 on 6 and 253 DF,  p-value: 0.99
```

The fitted models are:

$$\widehat{\textbf{Model I:}} \quad \widehat{\text{AvgTemp}}_t = 4.233 + 0.004\,t - 0.045\,\cos\left(\frac{2\pi t}{4}\right) - 0.096\,\sin\left(\frac{2\pi t}{4}\right),$$

$$\widehat{\textbf{Model II:}} \quad \widehat{\text{AvgTemp}}_t = 5952.734 - 5.998\,t + 0.003\,\frac{t^2}{2!} - 0.019\,\cos\left(\frac{2\pi t}{4}\right) - 0.078\,\sin\left(\frac{2\pi t}{4}\right),$$

$$\widehat{\textbf{Model III:}} \quad \widehat{\text{AvgTemp}}_t = 5914.417 - 5.960\,t + 0.003\,\frac{t^2}{2!} - 0.017\,\cos\left(\frac{2\pi t}{4}\right) - 0.077\,\sin\left(\frac{2\pi t}{4}\right) + 0.034\,\cos\left(\frac{4\pi t}{4}\right) + 0.137\,\sin\left(\frac{4\pi t}{4}\right).$$

where AvgTemp_t denotes the New Haven average temperature at time t, and $s = 4$ is the seasonal period from part (a).

Problem 2 (c)

Problem 2 (c)

Prepare a summary table with number of regression parameters, R_a^2, AIC, AICC and BIC. Which model best explains temperature changes and why?

Cell 15

```r
# Collect model-fit criteria.
fit_table <- function(fit) {
  n   <- length(resid(fit))
  r   <- length(coef(fit))
  sse <- sum(resid(fit)^2)
  data.frame(
    Parameters = r,
    R2a  = summary(fit)$adj.r.squared,
    AIC  = AIC(fit) / n,
    AICC = log(sse / n) + ((n + r) / (n - r - 2)),
    BIC  = AIC(fit, k = log(n)) / n
  )
}
comp <- dplyr::bind_rows(
  'Model I'   = fit_table(fit1),
  'Model II'  = fit_table(fit2),
  'Model III' = fit_table(fit3),
  .id = 'Model'
)
comp$Parameters <- as.integer(comp$Parameters)
comp <- comp %>%
  dplyr::mutate(
    R2a  = round(R2a, 3),
    AIC  = round(AIC, 3),
    AICC = round(AICC, 3),
    BIC  = round(BIC, 3)
  )
table_latex(comp, col_names = c('Model', '\\# Parameters', '$R_a^2$', 'AIC',
'AICC', 'BIC'))
```

Model	# Parameters	R_a^2	AIC	AICC	BIC
Model I	4	-0.012	7.158	5.321	7.226
Model II	5	-0.012	7.162	5.326	7.245
Model III	7	-0.020	7.178	5.342	7.287

From the comparison table:

- All three adjusted R^2 values are near zero (slightly negative), so none of these deterministic trend/seasonality models explains much variation in this New Haven quarterly-average series.

- Model I has the largest R_a^2 (least negative) among the three.
- Model I also has the smallest AIC, AICC, and BIC.

Therefore, **Model I** is selected: adding the quadratic and higher-harmonic terms does not improve fit enough to offset the complexity penalty.